Accelerated Algebra and Analysis: Analysis Part

Angus

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Abstract

This is a simple abstract for the Accelerated Algebra and Analysis notes it is focused on the analysis part.

1 Lecture 1

1.1 Introduction

Analysis asks not only how to calculate, but why the objects and rules exist. Familiar claims such as

$$\sqrt{2} \in \mathbb{R}, \qquad \frac{1}{n} \to 0, \qquad \frac{d}{dx}x^3 = 3x^2$$

require precise foundations: the existence of square roots, the meaning of convergence, and the reason differentiation rules hold. The starting point is that the real line has no gaps.

1.2 Upper Bounds and Supremum

Definition 1.1

Upper bound

Let $A \subseteq \mathbb{R}$. A number $M \in \mathbb{R}$ is an *upper bound* for A if

$$a \leq M \quad \text{for every } a \in A.$$

The set A is *bounded above* if

$$\exists M \in \mathbb{R};\ \forall a \in A,\ a \leq M.$$

Example

For the interval $(0, 1]$, the numbers $1, 2, 100$ are upper bounds, while $\frac{1}{2}$ is not.

Definition 1.2

Supremum

Let $A \subseteq \mathbb{R}$. A number $s \in \mathbb{R}$ is the *supremum* (least upper bound) of A if:

1. $a \leq s$ for every $a \in A$;
2. if M is any upper bound for A, then $s \leq M$.

We write $s = \sup A$.

Equivalently, $s = \sup A$ if s is an upper bound and every $x < s$ fails to be an upper bound; that is,

$$\forall x < s,\ \exists a \in A \colon x < a.$$

Theorem 1.1

Uniqueness of supremum

If a set has a supremum, then that supremum is unique.

Example

$\sup(0, 1] = 1$.

Proof. Clearly 1 is an upper bound for $(0, 1]$. Let M be any upper bound. If $M < 1$, then

$$a = \frac{M + 1}{2} \in (0, 1]$$

and $a > M$, contradicting that M is an upper bound. Hence $1 \leq M$. Therefore 1 is the least upper bound. □

Example

Let

$$A = \left\{ 1 - \frac{1}{n} \colon n \in \mathbb{N} \right\}.$$

Then $\sup A = 1$.

Proof. For every $n \in \mathbb{N}$,

$$1 - \frac{1}{n} < 1,$$

so 1 is an upper bound. Let $\epsilon > 0$. By the Archimedean property, choose $n \in \mathbb{N}$ with

$$\frac{1}{n} < \epsilon.$$

Then

$$1 - \frac{1}{n} > 1 - \epsilon.$$

Hence every number strictly below 1 fails to be an upper bound, so $\sup A = 1$. □

Definition 1.3

Maximum

A number M is the *maximum* of A if $M \in A$ and $a \leq M$ for every $a \in A$.

Remark

A supremum is a least upper bound; a maximum must also belong to the set. For example, $(0, 1]$ has maximum 1, while

$$\left\{1 - \frac{1}{n} : n \in \mathbb{N}\right\}$$

has supremum 1 but no maximum.

Remark

Two useful facts:

- If $\max A$ exists, then $\sup A$ exists and $\sup A = \max A$.
- If $\sup A$ exists and $\sup A \notin A$, then $\max A$ does not exist.

1.3 The Completeness Axiom

Theorem 1.2

Completeness Axiom

Every nonempty subset $A \subseteq \mathbb{R}$ that is bounded above has a least upper bound in $\mathbb{R}$.

Remark

Completeness is an axiom of the real number system. Arithmetic and order alone do not give it. For example, $\mathbb{Q}$ fails it. Consider

$$S = \{q \in \mathbb{Q} : q^2 < 2\},$$

which is nonempty and bounded above in $\mathbb{Q}$, but has no supremum in $\mathbb{Q}$.

Remark

Every hypothesis matters:

- The empty set is nonempty? No, so completeness does not apply.
- $[0, \infty)$ is not bounded above, so completeness does not apply.
- Replacing $\mathbb{R}$ by $\mathbb{Q}$ destroys completeness.

1.4 The Approximation Property

Theorem 1.3

Approximation property

Let s be an upper bound for $A \subseteq \mathbb{R}$. Then

$$s = \sup A \iff \forall \epsilon > 0,\ \exists a \in A\colon\ s - \epsilon < a \leq s.$$

Intuitively, every strip immediately to the left of s contains an element of A.

1.5 Lower Bounds and Infimum

Definition 1.4

Lower bound and infimum

Let $A \subseteq \mathbb{R}$.

- m is a *lower bound* for A if $m \leq a$ for every $a \in A$.
- A is *bounded below* if it has a lower bound; it is *bounded* if it is bounded both above and below.
- $t = \inf A$ if t is a lower bound and every lower bound m satisfies $m \leq t$.
- $m = \min A$ if $m \in A$ and $m \leq a$ for every $a \in A$.

Remark

Reflection rule: if A is nonempty and bounded below, then

$$\inf A = -\sup(-A), \qquad -A = \{-a : a \in A\}.$$

1.6 The Archimedean Property and Density

Theorem 1.4

Archimedean property

The natural numbers are not bounded above in $\mathbb{R}$. That is,

$$\forall x \in \mathbb{R},\ \exists n \in \mathbb{N}\colon\ n > x.$$

Equivalently, for every $r > 0$, there exists $n \in \mathbb{N}$ such that

$$\frac{1}{n} < r.$$

Definition 1.5

Dense set

A set $D \subseteq \mathbb{R}$ is *dense* in $\mathbb{R}$ if

$$\forall x \in \mathbb{R},\ \forall \epsilon > 0,\ \exists d \in D \colon |x - d| < \epsilon.$$

Equivalently, every open interval (u, v) with $u < v$ contains a point of D.

Theorem 1.5

Density of $\mathbb{Q}$

The rationals $\mathbb{Q}$ are dense in $\mathbb{R}$.

1.7 Existence of Square Roots

Theorem 1.6

Existence of square roots

For every $x > 0$, there exists a unique $s > 0$ such that

$$s^2 = x.$$

We denote this number by $\sqrt{x}$.

Remark

The proof uses completeness. For example, to show that there is a positive real number s with $s^2 = 2$, consider

$$S = \{r \geq 0 \colon r^2 < 2\}$$

and let $s = \sup S$. Completeness guarantees that s exists; one then proves $s^2 = 2$.

1.8 Sequences: Boundedness and Monotonicity

Definition 1.6

Sequence

A sequence is a function $a \colon \mathbb{N} \to \mathbb{R}$, written as $(a_n)_{n \in \mathbb{N}}$.

Definition 1.7

Boundedness of a sequence

- Bounded above means

$$\exists M \in \mathbb{R};\ \forall n \in \mathbb{N},\ a_n \leq M.$$

- Bounded below means

$$\exists L \in \mathbb{R};\ \forall n \in \mathbb{N},\ a_n \geq L.$$

- Bounded: bounded both above and below.

Definition 1.8

Monotonicity of a sequence

For every $n \in \mathbb{N}$:

a) Non-decreasing: $a_{n+1} \geq a_n$

b) Non-increasing: $a_{n+1} \leq a_n$

c) Increasing: $a_{n+1} > a_n$

d) Decreasing: $a_{n+1} < a_n$

A sequence is *monotone* if it is either non-decreasing or non-increasing.

Example

For each sequence, decide whether it is bounded above, bounded below, and monotone. Consider

$$a_n = 1 - \frac{1}{n}, \qquad b_n = \frac{1}{n}, \qquad c_n = (-1)^n, \qquad d_n = n.$$

- (a_n) is bounded, increasing, and converges to 1.
- (b_n) is bounded, decreasing, and converges to 0.
- (c_n) is bounded but not monotone.
- (d_n) is unbounded above, increasing.

2 Lecture 2

2.1 Three tools from last week

2.1.1 Completeness

Every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.

2.1.2 Approximation property

If $s = \sup A$, then for every $\epsilon > 0$, there exists an $a \in A$ such that

$$s - \epsilon < a \leq s.$$

2.1.3 Archimedean property

For every $r \in \mathbb{R}$ with $r > 0$ there exists $n \in \mathbb{N}$ such that

$$\frac{1}{n} < r.$$

2.2 Sequences

Definition 2.1

Sequence

A sequence is a function $a\colon \mathbb{N} \to \mathbb{R}$, written as $(a_n)_{n\in\mathbb{N}}$.

2.2.1 Boundedness

Definition 2.2

Boundedness

- Bounded above means

$$\exists M \in \mathbb{R};\ \forall n \in \mathbb{N},\ a_n \leq M.$$

- Bounded below means

$$\exists L \in \mathbb{R};\ \forall n \in \mathbb{N},\ a_n \geq L.$$

- Bounded: A sequence is bounded if it is both bounded above and bounded below.

2.2.2 Monotonicity

Definition 2.3

Monotonicity: for every $n \in \mathbb{N}$

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a) Non-decreasing: $a_{n+1} \geq a_n$

b) Non-increasing: $a_{n+1} \leq a_n$

c) Increasing: $a_{n+1} > a_n$

d) Decreasing: $a_{n+1} < a_n$

Monotone: A sequence is monotone if it is either non-decreasing or non-increasing.

2.2.3 Convergence

Definition 2.4

Convergence: ϵ–N Definition

A sequence $(a_n)_{n\in\mathbb{N}}$ converges to a limit $L \in \mathbb{R}$ if

$$\forall \epsilon > 0,\ \exists N \in \mathbb{N},\ \forall n \in \mathbb{N}\colon n > N \implies |a_n - L| < \epsilon.$$

Equivalently, for every $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that, for all $n > N$,

$$|a_n - L| < \epsilon.$$

What convergence means is being arbitrarily close to a limit, and eventually staying close to it.

$$a_n \to L \text{ or } \lim_{n\to\infty} a_n = L.$$

- The sequence converges if it has a real limit.
- If it has no real limit, it diverges.

Notation

∞ is not a real number, the definition contains only finite indices and real inequalities.

Example

Using the definition of convergence, prove that

$$a_n = \frac{1}{n} \to 0 \text{ as } n \to \infty.$$

Proof. Let $a_n = \frac{1}{n}$. For all $\epsilon > 0$, we need to find $N \in \mathbb{N}$ such that for all $n > N$,

$$|a_n - 0| < \epsilon.$$

Clearly,

$$|a_n - 0| = \left|\frac{1}{n} - 0\right| = \frac{1}{n}.$$

Choose

$$N = \left\lfloor \frac{1}{\epsilon} \right\rfloor + 1.$$

Then $N > \frac{1}{\epsilon}$, so $\frac{1}{N} < \epsilon$. For all $n > N$,

$$\frac{1}{n} < \frac{1}{N} < \epsilon.$$

Thus $|a_n - 0| < \epsilon$. Hence $a_n \to 0$ as $n \to \infty$. □

2.3 Convergence and Limitations

2.3.1 Uniqueness of Limits

Theorem 2.1

If a sequence $(a_n)_{n\in\mathbb{N}}$ converges to both a and b, $a, b \in \mathbb{R}$, then $a = b$.

Proof. Suppose $a_n \to a$ and $a_n \to b$. Let $\epsilon > 0$. Then there exist $N_1, N_2 \in \mathbb{N}$ such that

$$\forall n > N_1,\ |a_n - a| < \epsilon/2,$$
$$\forall n > N_2,\ |a_n - b| < \epsilon/2.$$

Let $m = \max\{N_1, N_2\}$. For all $n > m$,

$$|a - b| \leq |a - a_n| + |a_n - b| < \epsilon/2 + \epsilon/2 = \epsilon.$$

Since $\epsilon > 0$ was arbitrary, we have $a = b$. □

2.3.2 Limit laws

Theorem 2.2

Addition, scalar multiplication and multiplication laws

Suppose $a_n \to a$ and $b_n \to b$, with $c \in \mathbb{R}$. Then

$$a_n + b_n \to a + b,$$
$$ca_n \to ca,$$
$$a_n b_n \to ab.$$

For the sum, we have the following proof:

Proof. Let $\epsilon > 0$. Since $a_n \to a$, there exists $N_1 \in \mathbb{N}$ such that for all $n > N_1$,

$$|a_n - a| < \epsilon/2.$$

Since $b_n \to b$, there exists $N_2 \in \mathbb{N}$ such that for all $n > N_2$,

$$|b_n - b| < \epsilon/2.$$

Let $N = \max\{N_1, N_2\}$. Then for all $n > N$,

$$|(a_n + b_n) - (a + b)| = |(a_n - a) + (b_n - b)| \leq |a_n - a| + |b_n - b| < \epsilon/2 + \epsilon/2 = \epsilon.$$

Hence $a_n + b_n \to a + b$. □

Theorem 2.3

The quotient law

Suppose $a_n \to a$ and $b_n \to b$. If

$$b_n \neq 0 \text{ for every } n \in \mathbb{N}, \quad b \neq 0,$$

then

$$\frac{a_n}{b_n} \to \frac{a}{b}.$$

Notation

Check both denominator conditions: The quotients must be defined, and the limiting denominator must be nonzero.

2.3.3 The squeeze theorem

Theorem 2.4

The squeeze theorem

Suppose $l_n \leq a_n \leq u_n$ for all $n \in \mathbb{N}$, and

$$l_n \to a \text{ and } u_n \to a.$$

Then $a_n \to a$.

Example

Determine whether the sequence

$$a_n = \frac{\sin n}{n}$$

converges, and justify your answer.

2.3.4 Two Ways to Prove Divergence

Theorem 2.5

Boundedness

Every convergent real sequence is bounded.

But the converse is not true: a bounded sequence need not be convergent.

Example

Consider the sequence

$$a_n = (-1)^n.$$

This sequence is bounded since $|a_n| \leq 1$ for all n, but it does not converge because it oscillates between -1 and 1.

Definition 2.5

Subsequences

A subsequence is

$$(a_{n_k})_{k \in \mathbb{N}}$$

where $n_1 < n_2 < n_3 < \dots$ are natural numbers.

Theorem 2.6

Subsequence test

If $a_n \to a$, then every subsequence converges to a. Thus two subsequences with different limits imply divergence.

Example

Consider the sequence

$$a_n = (-1)^n$$

and the subsequences

$$a_{2k} = 1,$$
$$a_{2k-1} = -1.$$

This demonstrates the divergence of the original sequence.

2.4 Monotone Convergence and Bolzano-Weierstrass

2.4.1 Monotone Convergence

Theorem 2.7

Monotone Convergence Theorem

- If (a_n) is non-decreasing and bounded above, it converges to

$$\sup\{a_n : n \in \mathbb{N}\}.$$

- If (a_n) is non-increasing and bounded below, it converges to

$$\inf\{a_n : n \in \mathbb{N}\}.$$

The theorem provides the existence of a limit before its value is known. We prove the first part.

Proof. Assume (a_n) is non-decreasing and bounded above.
If $a_1 \leq a_2 \leq a_3 \leq \cdots \leq M$, then

$$a_n \to a = \sup \underbrace{\{a_1, a_2, a_3, \dots\}}_{\text{A}}.$$

For every $\epsilon > 0$, there exists $x \in A$ such that

$$a - \epsilon < x \leq a.$$

Replace $\exists x \in A$ with $\exists n \in \mathbb{N}$ such that

$$x = a_n.$$

Then

$$a - \epsilon < a_n \leq a_{n+1} \leq a_{n+2} \leq \cdots \leq a.$$

So for $k = n, n+1, n+2, \ldots$, we have

$$|a_k - a| < \epsilon.$$

Take $N = n$. Then for all $k \geq N$, $|a_k - a| < \epsilon$, which proves $a_n \to a$. □

2.4.2 Bolzano-Weierstrass Theorem

Theorem 2.8

Every bounded sequence of real numbers has a convergent subsequence.

Lemma 2.9

Lemma used in the proof

Every real sequence has a monotone subsequence.

The full sequence need not converge.

Example

Apply Bolzano-Weierstrass Theorem to each sequence

$$a_n = (-1)^n, \quad b_n = \sin n, \quad c_n = n.$$

2.5 Cauchy Sequences and Divergence

2.5.1 Cauchy sequences

Definition 2.6

A real sequence (a_n) is Cauchy if

$$\forall \epsilon > 0,\ \exists N \in \mathbb{N},\ \forall m, n > N \colon |a_m - a_n| < \epsilon.$$

The definition compares late terms with one another, without specifying a limit.

Cauchy criterion in $\mathbb{R}$

A real sequence is convergent if and only if it is Cauchy.

2.5.2 Divergence to infinity

Definition 2.7

Divergence to infinity

$$a_n \to +\infty \iff \forall M \in \mathbb{R},\ \exists N \in \mathbb{N},\ \forall n > N \colon a_n > M,$$
$$a_n \to -\infty \iff \forall M \in \mathbb{R},\ \exists N \in \mathbb{N},\ \forall n > N \colon a_n < M.$$

These are types of divergence.

2.5.3 Useful elementary limits

Standard limits
For $p > 0$ and $r \in \mathbb{R}$ with $|r| < 1$,

$$\frac{1}{n^p} \to 0, \qquad r^n \to 0.$$

Theorem 2.10

Order and limits

$$a_n \leq b_n \text{ for all sufficiently large } n, \quad a_n \to a, \quad b_n \to b \implies a \leq b.$$

3 Lecture 3

3.1 Series

3.1.1 Definition of series

Difference between series and sequences: A series is the sum of the terms of a sequence. For sequences, convergence is defined as follows.

Definition 3.1

$(a_n)_{n\in\mathbb{N}}$ is convergent to a if

$$\forall \epsilon > 0,\ \exists N \in \mathbb{N},\ \forall n > N \colon |a_n - a| < \epsilon.$$

For series, convergence is defined in terms of the convergence of the sequence of partial sums.

Definition 3.2

We say

$$S \in \mathbb{R}$$

is the sum of the series

$$\sum_{n=1}^{\infty} a_n$$

if the sequence of partial sums satisfies

$$S_m \to S,$$

that is,

$$S = \lim_{m\to\infty} S_m.$$

Theorem 3.1

The series

$$S = \sum_{n=1}^{\infty} a_n$$

is convergent if the sequence of partial sums

$$S_m = \sum_{n=1}^{m} a_n = a_1 + a_2 + \cdots + a_m$$

is convergent as $m \to \infty$.

If the sequence of partial sums

$$(S_m)_{m \in \mathbb{N}}$$

has no real limit, the series is said to be divergent.

Example

For

$$a_k = 2^{-k}, \quad k \in \mathbb{N},$$

the sequence is

$$a_1, a_2, a_3, \cdots = 2^{-1}, 2^{-2}, 2^{-3}, \ldots$$

The partial sums are

$$\begin{aligned} S_1 &= a_1, \\ S_2 &= a_1 + a_2, \\ S_3 &= a_1 + a_2 + a_3, \\ &\vdots \\ S_m &= 1 - 2^{-m} \to 1. \end{aligned}$$

Therefore,

$$\sum_{k=1}^{\infty} a_k = 1.$$

Example

$a_k = 1,\ a_k = (-1)^k, \quad k \in \mathbb{N}$

$$\sum_{k=1}^{\infty} 1 = 1 + 1 + 1 + \ldots \text{ diverges},$$

$$\sum_{k=1}^{\infty} (-1)^k = -1 + 1 - 1 + 1 - \ldots \text{ diverges}.$$

3.1.2 Geometric series

Definition 3.3

A series with terms cr^k is geometric.

Theorem 3.2

The geometric series

$$\sum_{k=0}^{\infty} cr^k$$

converges if $|r| < 1$ and its sum is

$$\sum_{k=0}^{\infty} cr^k = \frac{c}{1-r}.$$

If $|r| \geq 1$, the series diverges.

Proof.

$$\begin{aligned} S_m = \sum_{k=0}^{m} cr^k &= c + cr + cr^2 + \cdots + cr^m \\ rS_m &= cr + cr^2 + \cdots + cr^{m+1} \\ S_m - rS_m &= c - cr^{m+1} \\ S_m(1-r) &= c(1-r^{m+1}) \\ S_m &= \frac{c(1-r^{m+1})}{1-r} \end{aligned}$$

For $|r| < 1$,

$$\sum_{k=0}^{\infty} cr^k = \frac{c(1-r^{m+1})}{1-r} \to \frac{c}{1-r}.$$

For $|r| = 1$,

$$\sum_{k=0}^{\infty} 1 = 1 + 1 + 1 + \ldots \text{ diverges,}$$

$$\sum_{k=0}^{\infty} (-1)^k = 1 - 1 + 1 - 1 + \ldots \text{ diverges.}$$

For $|r| > 1$, assume $S_n \to S$. Then

$$\begin{aligned} r^{m+1} &= S_{m+1} - S_m \\ &\to S - S = 0, \end{aligned}$$

which is false since $r^{m+1} \nrightarrow 0$. □

3.1.3 Telescoping series

Definition 3.4

A series is telescoping if its terms can be written as

$$a_k = b_k - b_{k+1}, \quad k \in \mathbb{N},$$

for some sequence $(b_k)_{k\in\mathbb{N}}$.

Remark

$$\begin{aligned} S_m &= \sum_{k=1}^{m} a_k \\ &= \sum_{k=1}^{m} (b_k - b_{k+1}) = b_1 - b_{m+1}. \end{aligned}$$

S_m converges $\Leftrightarrow$ (b_n) converges.

3.2 Convergence

3.2.1 The kth-term test

Theorem 3.3

If

$$\sum_{k=1}^{\infty} a_k \text{ converges, then } a_k \to 0.$$

Proof. If $\sum a_k$ converges, write

$$S_m = \sum_{k=1}^{m} a_k$$

and suppose $S_m \to S$. Then

$$a_{m+1} = S_{m+1} - S_m \to S - S = 0.$$

Hence $a_k \to 0$. □

But if $a_k \to 0$, does not mean the series converges.
For example, the harmonic series $\sum_{k=1}^{\infty} \frac{1}{k}$ has terms tending to 0, but the series diverges.

3.2.2 The harmonic series

Theorem 3.4

The harmonic series

$$\sum_{k=1}^{\infty} \frac{1}{k}$$

diverges.

But its terms tend to 0.

Proof. Consider the partial sums

$$S_m = \sum_{k=1}^{m} \frac{1}{k}.$$

We have

$$S_{2^n} = 1 + \underbrace{\frac{1}{2}}_{\geq \frac{1}{2}} + \underbrace{\frac{1}{3} + \frac{1}{4}}_{\geq \frac{1}{2}} + \underbrace{\frac{1}{5} + \cdots + \frac{1}{8}}_{\geq \frac{1}{2}} + \cdots + \underbrace{\frac{1}{2^{n-1}+1} + \cdots + \frac{1}{2^n}}_{\geq \frac{1}{2}}.$$

Each group contributes at least $\frac{1}{2}$, so $S_{2^n} \to \infty$ as $n \to \infty$. Hence, the harmonic series diverges. □

3.2.3 The boundedness test for non-negative series

Theorem 3.5

The boundedness test for non-negative series

Suppose $a_k \geq 0$ for all $k \in \mathbb{N}$ and $s_n = \sum_{k=1}^{n} a_k$. Then

$$\sum_{k=1}^{\infty} a_k \text{ converges } \Leftrightarrow (s_n) \text{ is bounded.}$$

Proof. Claim: $(s_n)_{n\in\mathbb{N}}$ is monotone nondecreasing.

$$s_{m+1} - s_m = a_{m+1} \geq 0 \implies s_{m+1} \geq s_m.$$

Therefore, if (s_n) is bounded, it must converge, and hence the series converges. Conversely, if the series converges, the partial sums (s_n) are bounded. □

3.2.4 The comparison test

Theorem 3.6

The comparison test

Suppose $0 \leq a_k \leq b_k$ for all $k \in \mathbb{N}$. Then

$$\sum_{k=1}^{\infty} b_k \text{ converges } \implies \sum_{k=1}^{\infty} a_k \text{ converges.}$$

Proof. Let $0 \leq a_k \leq b_k$ for all $k \in \mathbb{N}$, and suppose $\sum_{k=1}^{\infty} b_k$ converges. Then there exists $M \in \mathbb{R}$ such that

$$\sum_{k=1}^{n} b_k \leq M \quad \text{for all } n \in \mathbb{N}.$$

Since $0 \leq a_k \leq b_k$, we have

$$\sum_{k=1}^{n} a_k \leq \sum_{k=1}^{n} b_k \leq M \quad \text{for all } n \in \mathbb{N}.$$

Therefore, $\sum_{k=1}^{\infty} a_k$ is bounded and hence converges. □

Example

Consider the series

$$\sum_{k=1}^{\infty} \frac{k+1}{k^3+2}.$$

We have

$$0 \leq \frac{k+1}{k^3+2} \leq \frac{2k}{k^3} = \frac{2}{k^2} \leq \frac{2}{k(k-1)}.$$

Also,

$$\sum_{k=2}^{\infty} \frac{2}{k(k-1)} = 2 < \infty,$$

so

$$\sum_{k=1}^{\infty} \frac{k+1}{k^3+2} \text{ converges.}$$

3.2.5 The condensation test

Theorem 3.7

Suppose a_k is non-negative and non-increasing. Then the series $\sum_{k=1}^{\infty} a_k$ converges if and only if the series $\sum_{j=0}^{\infty} 2^j a_{2^j}$ converges, in the sense that

$$\sum_{k=1}^{\infty} a_k \text{ converges } \Leftrightarrow \sum_{j=0}^{\infty} 2^j a_{2^j} \text{ converges.}$$

The condensed terms sample the original series at indices that are powers of 2.

Proof.

$$S_m = \sum_{k=1}^{m} a_k,$$

$$S_{2^m} = a_1 + a_2 + \underbrace{(a_3 + a_4)}_{\geq 2a_4} + \underbrace{(a_5 + \cdots + a_8)}_{\geq 4a_8} + \cdots + \underbrace{(a_{2^{m-1}+1} + \cdots + a_{2^m})}_{\geq 2^{m-1}a_{2^m}},$$

$$2M \geq 2S_{2^m} \geq 2a_1 + 2a_2 + 2^2 a_{2^2} + \cdots + 2^m a_{2^m} \geq a_1 + \sum_{j=1}^{m} 2^j a_{2^j}.$$

□

3.2.6 The p-test

Theorem 3.8

p test

For $p \in \mathbb{R}$, the series $\sum_{k=1}^{\infty} \frac{1}{k^p}$ converges $\iff$ $p > 1$.

Proof. Let

$$a_k = \frac{1}{k^p}.$$

- If $p \leq 0$, then $\frac{1}{k^p} \geq 1$, so the terms do not tend to zero and the series diverges.
- If $p > 0$, then $a_k = \frac{1}{k^p}$ is non-negative and non-increasing, so the condensation test applies

$$\sum_{j=1}^{\infty} 2^j a_{2^j} = \sum_{j=1}^{\infty} 2^j \frac{1}{(2^j)^p} = \sum_{j=1}^{\infty} (2^{1-p})^j.$$

This geometric series converges exactly when $2^{1-p} < 1$, which holds if and only if $p > 1$.
Therefore, by the condensation test, the original series converges if and only if $p > 1$.

□

3.2.7 The ratio test

Theorem 3.9

The ratio test

Let $a_k > 0$ for all $k \in \mathbb{N}$ and suppose

$$r = \lim_{k \to \infty} \frac{a_{k+1}}{a_k}.$$

Then

- if $r < 1$, $\sum_{k=1}^{\infty} a_k$ converges;
- if $r > 1$, $\sum_{k=1}^{\infty} a_k$ diverges;
- if $r = 1$, the test is inconclusive.

Proof. If $r < 1$, choose s such that $r < s < 1$. Then, for all $k > N$,

$$\frac{a_{k+1}}{a_k} < s.$$

Then for all $k > N$,

$$0 \leq a_{N+k} \leq s^k a_N.$$

The right-hand side is a geometric series with ratio $s < 1$, which converges.

$$\sum_{k=1}^{\infty} a_{N+k} \leq \sum_{k=1}^{\infty} s^k a_N < \infty.$$

So

$$\sum_{k=1}^{\infty} a_{N+k}$$

is convergent. Therefore, the original series $\sum_{k=1}^{\infty} a_k$ also converges. □

Example

Determine whether each series converges or diverges.

$$\sum_{k=1}^{\infty} \frac{k}{k^2}, \quad \sum_{k=1}^{\infty} \frac{k!}{k^k}.$$

3.2.8 The alternating series test

Theorem 3.10

The alternating series test

Suppose (b_k) satisfies

- $b_k \geq 0$ for all $k \in \mathbb{N}$,
- b_k is non-increasing, i.e., $b_{k+1} \leq b_k$ for all k,
- $b_k \to 0$.

Then the alternating series

$$\sum_{k=1}^{\infty} (-1)^k b_k$$

converges.

Example

Consider the series

$$\sum_{k=1}^{\infty} \frac{(-1)^k}{k}.$$

3.2.9 Absolute and conditional convergence

Definition 3.5

A real series $\sum a_k$ converges absolutely if

$$\sum_{k=1}^{\infty} |a_k| \text{ converges.}$$

It converges conditionally if

$$\sum_{k=1}^{\infty} |a_k| \text{ diverges but } \sum_{k=1}^{\infty} a_k \text{ converges.}$$

To claim conditional convergence, both assertions must be established.

Theorem 3.11

The absolute convergence test

Every absolutely convergent real series is convergent. Indeed,

$$\sum_{k=1}^{\infty} |a_k| \text{ converges} \implies \sum_{k=1}^{\infty} a_k \text{ converges.}$$

Proof. Let $\sum_{k=1}^{\infty} |a_k|$ converge.

$$0 \le b_k = a_k + |a_k| \le 2|a_k|.$$

Since

$$a_k = b_k - |a_k|,$$

we have

$$\sum_{k=1}^{\infty} b_k \text{ converges by the comparison test.}$$

Also,

$$S_m = \sum_{k=1}^{m} a_k = \sum_{k=1}^{m} (b_k - |a_k|) = \sum_{k=1}^{m} b_k - \sum_{k=1}^{m} |a_k|.$$

Thus, S_m converges as $m \to \infty$. □

Example

Classify the type of convergence for the series

$$\sum_{k=1}^{\infty} \frac{\sin k}{k^2}, \quad \sum_{k=1}^{\infty} \frac{(-1)^k}{k}.$$

1. Conditionally convergent means

$$\sum_{k=1}^{\infty} \frac{(-1)^k}{k} \text{ converges by the alternating series test, but}$$
$$\sum_{k=1}^{\infty} \left| \frac{(-1)^k}{k} \right| = \sum_{k=1}^{\infty} \frac{1}{k} \text{ diverges.}$$

2. Absolutely convergent means

$$0 \le \left| \frac{\sin k}{k^2} \right| \le \frac{1}{k^2},$$

and

$$\sum_{k=1}^{\infty} \frac{1}{k^2} \text{ converges by the } p\text{-series test with } p = 2.$$

3.3 Summary of Convergence Tests

What you notice	A useful first check
Terms do not tend to zero	kth-term test
Non-negative algebraic terms	comparison; p-series
Factorials or exponential powers	ratio test
Non-increasing non-negative terms	condensation
Alternating signs	all three alternating-test hypotheses
General signed terms	absolute convergence first

Example

Choose a test and decide convergence for

$$\sum_{k=1}^{\infty} \frac{k}{3^k} \quad \text{and} \quad \sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{k}}.$$

4 Lecture 4

Continuity and limits

4.1 Recall

In this section, we study the continuity and limits of functions on the real line, which is similar to the sequences studied earlier.

For every $N \in \mathbb{N}$, there exists an $\epsilon > 0$ such that for all $n \geq N$,

$$|a_n - a| < \epsilon.$$

choose the accuracy ϵ first and find the corresponding N.

4.2 Functions on the Real Line

Definition 4.1

Domain and image of a function

For $f\colon I \to \mathbb{R}$, I is the domain of f and

$$f(I) = \{f(x) \mid x \in I\}$$

is called the image of f.

Definition 4.2

Composition of functions

For $f\colon I \to \mathbb{R}$ and $g\colon J \to \mathbb{R}$ with $f(I) \subseteq J$, the composition $g \circ f\colon I \to \mathbb{R}$ is defined by

$$(g \circ f)(x) = g(f(x)) \quad \text{for all } x \in I.$$

4.3 Bounded and Monotone Functions

Definition 4.3

Bounded function

A function $f: I \to \mathbb{R}$

1. Bounded above: $\exists M > 0$ such that $f(x) \leq M$ for all $x \in I$.
2. Bounded below: $\exists m > 0$ such that $f(x) \geq m$ for all $x \in I$.
3. Bounded: f is bounded if it is both bounded above and bounded below.

Definition 4.4

Monotonicity

A function $f: I \to \mathbb{R}$, for every $x_1, x_2 \in I$ with $x_1 < x_2$,

- f is non-decreasing if $f(x_1) \leq f(x_2)$ for all $x_1, x_2 \in I$.
- f is non-increasing if $f(x_1) \geq f(x_2)$ for all $x_1, x_2 \in I$.
- f is increasing if $f(x_1) < f(x_2)$ for all $x_1, x_2 \in I$.
- f is decreasing if $f(x_1) > f(x_2)$ for all $x_1, x_2 \in I$.

Monotone functions are either non-decreasing or non-increasing on their domain.

4.4 Continuity

4.4.1 Continuity at a Point

Definition 4.5

Continuity at a point

Let $I \subseteq \mathbb{R}$ be an interval and $a \in I$. A function $f: I \to \mathbb{R}$ is continuous at a if

$$\forall \epsilon > 0,\ \exists \delta > 0,\ \forall x \in I: |x - a| < \delta \implies |f(x) - f(a)| < \epsilon.$$

- ϵ specifies the output accuracy.
- δ specifies the input accuracy corresponding to the desired output accuracy.
- The definition ensures that small changes in the input x around a result in small changes in the output $f(x)$ around $f(a)$.

Example

Using the ϵ-δ definition, prove that

$$f(x) = 3x - 1$$

is continuous at every point $a \in \mathbb{R}$.

Proof. Pick $\epsilon > 0$, then

$$|f(x) - f(a)| = |3x - 1 - (3a - 1)| = |3(x - a)| = 3|x - a|.$$

To make $|f(x) - f(a)| < \epsilon$, choose $\delta = \frac{\epsilon}{3}$. Then if $|x - a| < \delta$, we have

$$|f(x) - f(a)| = 3|x - a| < 3\delta = \epsilon.$$

□

Example

Using the ϵ-δ definition, prove that

$$f(x) = x^3$$

is continuous at $a = 1$.

Proof. Pick $\epsilon > 0$, then

$$|f(x) - f(1)| = |x^3 - 1^3| = |x - 1||x^2 + x + 1|.$$

To estimate the second factor, first restrict $|x - 1| < 1$. Then $0 < x < 2$, so

$$|x^2 + x + 1| = x^2 + x + 1 < 4 + 2 + 1 = 7.$$

Hence

$$|f(x) - f(1)| \leq 7|x - 1|.$$

To make $|f(x) - f(1)| < \epsilon$, it is enough to require

$$7|x - 1| < \epsilon,$$

that is,

$$|x - 1| < \frac{\epsilon}{7}.$$

Therefore, choose

$$\delta = \min\left\{1, \frac{\epsilon}{7}\right\}.$$

Then if $|x - 1| < \delta$, we have

$$|f(x) - f(1)| = |x^3 - 1^3| = |x - 1||x^2 + x + 1| \leq 7|x - 1| < 7\delta \leq \epsilon.$$

Function $f(x) = x^3$ is therefore continuous at $a = 1$. □

Remark

The a cube minus b cube can be factored as

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2),$$

which is useful in ϵ-δ proofs for cubic functions.

4.4.2 Discontinuity at a Point

Definition 4.6

f is not continuous at a

$$\exists \epsilon > 0,\ \forall \delta > 0,\ \exists x \in I\colon |x - a| < \delta \text{ and } |f(x) - f(a)| \geq \epsilon.$$

Example

Let $f\colon \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} 0 & \text{if } x < 0, \\ 1 & \text{if } x \geq 0. \end{cases}$$

Then f is not continuous at $a = 0$.

Proof. Take $\epsilon = \frac{1}{2}$. Then for any $\delta > 0$, choose $x = -\frac{\delta}{2}$, we have

$$|x - 0| = \frac{\delta}{2} < \delta,$$

but

$$|f(x) - f(0)| = |f(-\frac{\delta}{2}) - f(0)| = |0 - 1| = 1 \geq \epsilon.$$

Therefore, f is not continuous at $a = 0$. □

4.4.3 Continuity on an Interval

Definition 4.7

Continuity on an interval

A function $f\colon I \to \mathbb{R}$ is continuous on I if it is continuous at every point $x_0 \in I$.

$$\forall a \in I,\ \forall \epsilon > 0,\ \exists \delta > 0,\ \forall x \in I\colon |x - a| < \delta \implies |f(x) - f(a)| < \epsilon.$$

The δ can depend on both ϵ and the point $a \in I$.

Remark

The interval must be nondegenerate, meaning it contains more than one point.

Example

for $g\colon [0,+\infty) \to \mathbb{R}$ defined by

$$g(x) = 1,$$

we have g is continuous on $[0,+\infty)$.

Proof. Let $a \in [0,+\infty)$ and $\epsilon > 0$ be given. Choose $\delta = \epsilon$. Then for any $x \in [0,+\infty)$ with $|x-a| < \delta$, we have

$$|g(x) - g(a)| = |1-1| = 0 < \epsilon.$$

Therefore, g is continuous at every point $a \in [0,+\infty)$, and hence continuous on $[0,+\infty)$. □

The domain plays the role in determining the points at which a function needs to be continuous.

For example, a function may be continuous on its domain even if it is not continuous everywhere on the real line.

4.4.4 Laws for Continuous Functions

Theorem 4.1

Sums, products, and quotients of continuous functions are continuous.

More precisely, if f and g are continuous at a, then

$$f+g, \quad f-g, \quad f \cdot g \text{ are continuous at } a,$$

and if $g(a) \neq 0$, then

$$\frac{f}{g} \text{ is continuous at } a.$$

Theorem 4.2

Composition

If $f\colon I \to J$ is continuous at a and $g\colon J \to \mathbb{R}$ is continuous at $f(a)$, then the composition $g \circ f$ is continuous at a.

Remark

Polynomial Functions are Continuous on $\mathbb{R}$, Rational Functions are Continuous where their denominators are non-zero.

Proof. For a polynomial function

$$f(x) = \sum_{k=0}^{n} c_k x^k,$$

Each term $c_k x^k$ is continuous, and sums of continuous functions are continuous. □

Example

Explain where this function is continuous.

$$h(x) = \sin(x^3)$$

Proof. Pick a, and let $\epsilon > 0$ be given.

$$|h(x) - h(a)| = |\sin(x^3) - \sin(a^3)| = |(\sin\xi)'||x^3 - a^3| \leq |x^3 - a^3| < \epsilon,$$

which shows that h is continuous at a. □

4.4.5 Sequential Characterization of Continuity

Theorem 4.3

Sequential Characterization of Continuity

Let $f\colon I \to \mathbb{R}$ be continuous at $a \in I$. If

$$a_n \in I, \text{ for all } n, \quad a_n \to a,$$

then

$$f(a_n) \to f(a).$$

4.4.6 Local Consequence of Continuity

Lemma 4.4

Strict inequalities persist locally

Let $f\colon I \to \mathbb{R}$ be continuous at $c \in I$ and $y \in \mathbb{R}$

- If $f(c) > y$, then there is $\delta > 0$ such that

$$x \in I, \ |x - c| < \delta \implies f(x) > y.$$

- If $f(c) < y$, then there is $\delta > 0$ such that

$$x \in I, \ |x - c| < \delta \implies f(x) < y.$$

4.5 Limits of Functions

4.5.1 Limit of a Function at a Point

Definition 4.8

Let I be a non-degenerate interval, $a \in I$, and let f be defined on $I \setminus \{a\}$.
We say that the limit of $f(x)$ as x approaches a is L, and write

$$\lim_{x \to a} f(x) = L$$

if

$$\forall \epsilon > 0,\ \exists \delta > 0,\ \forall x \in I \setminus \{a\} \colon 0 < |x - a| < \delta \implies |f(x) - L| < \epsilon.$$

Remark

The limit with continuity is related in the sense that if a function f is continuous at a point a, then

$$\lim_{x \to a} f(x) = f(a).$$

If we have a non-continuous function at a but there exists a limit L, we can complete the function to a continuous function by defining

$$\tilde{f}(x) = \begin{cases} f(x), & x \neq a, \\ L, & x = a. \end{cases}$$

Example

Find and justify

$$\lim_{x \to 1} \frac{x^2 - 1}{x - 1}.$$

The displayed function is defined for all $x \neq 1$.

Proof. We can factor the numerator as $x^2 - 1 = (x - 1)(x + 1)$. Then for $x \neq 1$,

$$\frac{x^2 - 1}{x - 1} = \frac{(x - 1)(x + 1)}{x - 1} = x + 1.$$

Therefore,

$$\lim_{x \to 1} \frac{x^2 - 1}{x - 1} = \lim_{x \to 1} (x + 1) = 2.$$

□

And the function can be completed to a continuous function at $x = 1$ by defining

$$\tilde{f}(x) = \begin{cases} \frac{x^2 - 1}{x - 1}, & x \neq 1, \\ 2, & x = 1. \end{cases}$$

Remark

Consider

$$\tilde{f}(x) = \begin{cases} \frac{x^2-1}{x-1}, & x \neq 1, \\ c, & x = 1. \end{cases}$$

c is an arbitrary constant; it does not affect the limit as $x \to 1$. The limit is determined solely by the behavior of the function for $x \neq 1$, which is

$$\lim_{x \to 1} \frac{x^2 - 1}{x - 1} = 2.$$

Theorem 4.5

Let $f \colon I \to \mathbb{R}$ be a function defined on an interval I, and let $a \in I$. If I is a non-degenerate interval around a, then

$$\lim_{x \to a} f(x) = f(a) \iff f \text{ is continuous at } a.$$

4.5.2 Limit Law for Functions

Theorem 4.6

Suppose f and g are functions defined on an interval I and let $a \in I$. If

$$\lim_{x \to a} f(x) = a_f \quad \text{and} \quad \lim_{x \to a} g(x) = a_g,$$

then

- addition
$$\lim_{x \to a} (f(x) + g(x)) = a_f + a_g;$$
- subtraction
$$\lim_{x \to a} (f(x) - g(x)) = a_f - a_g;$$
- multiplication
$$\lim_{x \to a} (f(x) \cdot g(x)) = a_f \cdot a_g;$$
- division
$$\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{a_f}{a_g}, \quad a_g \neq 0.$$

4.6 Properties of Continuous Functions on Closed Intervals

4.6.1 The Intermediate Value Theorem

Theorem 4.7

Intermediate Value Theorem (IVT)

Let $a < b$ and $f\colon [a,b] \to \mathbb{R}$ be continuous. With $f(a) \le f(b)$, for any y with

$$f(a) \le y \le f(b),$$

there exists $c \in [a,b]$ such that $f(c) = y$.

Proof.

$$S = \{x \in [a,b] \mid f(x) \le y\}.$$

Since

$$f(a) \le y \implies a \in S,$$

S is nonempty; let

$$c = \sup S.$$

By the continuity of f, we have

$$f(c) = y.$$

□

Example

Use the IVT to show that there is $s \in (1,2)$ such that $s^2 = 2$.

$$f(x) = x^2, \quad f(1) = 1, \quad f(2) = 4.$$

By the IVT, since $2 \in (f(1), f(2)) = (1,4)$, there exists $s \in (1,2)$ such that $f(s) = 2$.

4.6.2 Boundedness Theorem

Theorem 4.8

Boundedness Theorem

Let $f\colon [a,b] \to \mathbb{R}$ be continuous. Then f is bounded; that is, there exists $M > 0$ such that

$$|f(x)| \le M \quad \text{for all } x \in [a,b].$$

Proof. Suppose that f is unbounded above. Then for each $n \in \mathbb{N}$, there exists $x_n \in [a,b]$ such that

$$f(x_n) \ge n.$$

By the Bolzano–Weierstrass theorem, the sequence (x_n) has a convergent subsequence

$$x_{n_k} \to c \in [a,b].$$

By the continuity of f, we have

$$f(x_{n_k}) \to f(c).$$

But

$$f(x_{n_k}) \geq n_k,$$

so

$$f(x_{n_k}) \to \infty.$$

This is a contradiction, hence f is bounded above. A similar argument shows that f is bounded below, hence f is bounded. □

4.6.3 Extreme Value Theorem

Theorem 4.9

Extreme Value Theorem (EVT)

Let $f\colon [a,b] \to \mathbb{R}$ be continuous and $a < b$. Then there exist $x_m, x_M \in [a,b]$ such that

$$f(x_m) \leq f(x) \leq f(x_M) \quad \text{for all } x \in [a,b].$$

It can be proven using the Boundedness Theorem and the fact that a continuous function on a closed interval attains its bounds.

5 Lecture 5

Differentiability and the mean value theorem.

5.1 The Derivative

5.1.1 Derivative and tangent line

Definition 5.1

Let $I \subseteq \mathbb{R}$ be an interval and $f\colon I \to \mathbb{R}$ a function. We say that f is *differentiable* at $c \in I$ if the following limit exists.

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c}.$$

This number $f'(c)$ is called the *derivative* of f at c.

Remark

If f is differentiable at $c \in I$, we say that f is differentiable at c. And f' is the derivative function defined on I by $f'(c)$ for each $c \in I$.

Definition 5.2

The tangent line to the graph of f at $c \in I$ is the line passing through the point $(c, f(c))$ with slope $f'(c)$. Its equation is

$$y = f(c) + f'(c)(x - c).$$

Proof. The line passing through $(c, f(c))$ and $(c+h, f(c+h))$ for some $h \neq 0$ has slope

$$\frac{f(c+h)-f(c)}{h}.$$

Taking the limit as $h \to 0$, we obtain the slope of the tangent line:

$$f'(c) = \lim_{h\to 0} \frac{f(c+h)-f(c)}{h}.$$

Hence, the equation of the tangent line is

$$y = f(c) + f'(c)(x-c).$$

□

Theorem 5.1

Standard derivatives

$$\begin{aligned} &\frac{d}{dx}c = 0 && \frac{d}{dx}\cos x = -\sin x \\ &\frac{d}{dx}x^n = nx^{n-1},\ n \in \mathbb{N} && \frac{d}{dx}e^x = e^x \\ &\frac{d}{dx}\sin x = \cos x && \frac{d}{dx}\ln x = \frac{1}{x},\ x > 0 \end{aligned}$$

Example

Let $f(x) = \frac{1}{x}$, $x \neq 0$. Then find $f'(2)$.

We have (for $h \neq -2, h \neq 0$),

$$\begin{aligned} f'(2) &= \lim_{h\to 0} \frac{f(2+h)-f(2)}{h} \\ &= \lim_{h\to 0} \frac{\frac{1}{2+h} - \frac{1}{2}}{h} \\ &= \lim_{h\to 0} \frac{2-(2+h)}{2(2+h)h} \\ &= \lim_{h\to 0} -\frac{1}{2(2+h)} \\ &= -\frac{1}{4}. \end{aligned}$$

5.1.2 Differentiability Implies Continuity

Theorem 5.2

Let I be an open interval and $f : I \to \mathbb{R}$ be differentiable at $c \in I$. Then f is continuous at c.

Remark

Converse is not necessarily true; a function can be continuous at a point but not differentiable there.

Proof. Since f is differentiable at c, we have

$$f'(c) = \lim_{h \to 0} \frac{f(c+h) - f(c)}{h}.$$

Letting $h \to 0$, we get

$$\begin{aligned} \lim_{h \to 0}(f(c+h) - f(c)) &= \lim_{h \to 0} \frac{f(c+h) - f(c)}{h} \cdot \lim_{h \to 0} h \\ &= f'(c) \cdot 0 \\ &= 0. \end{aligned}$$

Hence,

$$\lim_{x \to c} f(x) = f(c),$$

which shows that f is continuous at c. □

5.1.3 Differentiation laws

Theorem 5.3

Linearity and product rule of differentiation hold.

For differentiable functions f and g and a constant c,

$$(\alpha f + \beta g)'(a) = \alpha f'(a) + \beta g'(a),$$

And

$$(fg)'(a) = f'(a)g(a) + f(a)g'(a).$$

Proof. The product rule. Let $h \to 0$. We have

$$\begin{aligned} (fg)'(a) &= \lim_{h \to 0} \frac{f(a+h)g(a+h) - f(a)g(a)}{h} \\ &= \lim_{h \to 0} \frac{f(a+h)g(a+h) - f(a)g(a+h) + f(a)g(a+h) - f(a)g(a)}{h} \\ &= \lim_{h \to 0} \frac{(f(a+h) - f(a))g(a+h) + f(a)(g(a+h) - g(a))}{h} \\ &= \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} g(a+h) + f(a) \lim_{h \to 0} \frac{g(a+h) - g(a)}{h} \\ &= f'(a)g(a) + f(a)g'(a). \end{aligned}$$

The linearity of differentiation can be proved similarly by considering

$$(\alpha f + \beta g)'(a) = \lim_{h \to 0} \frac{\alpha f(a+h) + \beta g(a+h) - (\alpha f(a) + \beta g(a))}{h} = \alpha f'(a) + \beta g'(a).$$

□

Theorem 5.4

Quotient rule.

For differentiable functions f and g with $g(a) \neq 0$,

$$\left(\frac{f}{g}\right)'(a) = \frac{f'(a)g(a) - f(a)g'(a)}{(g(a))^2}.$$

Remark

$$g(a) \neq 0 \implies \exists \delta > 0 \text{ such that } |h| < \delta \implies g(a+h) \neq 0.$$

Theorem 5.5

Chain rule.

For differentiable functions f and g,

$$(f \circ g)'(a) = f'(g(a)) \cdot g'(a).$$

Proof. Let $h \to 0$. We have

$$\begin{aligned}(f \circ g)'(a) &= \lim_{h \to 0} \frac{f(g(a+h)) - f(g(a))}{h} \\ &= \lim_{h \to 0} \frac{f(g(a+h)) - f(g(a))}{g(a+h) - g(a)} \cdot \frac{g(a+h) - g(a)}{h} \\ &= f'(g(a)) \cdot g'(a).\end{aligned}$$

□

5.2 Local Extrema and the mean value theorem

5.2.1 Local Extrema and Stationary points

Definition 5.3

Let $f \colon I \to \mathbb{R}$ be a function defined on an interval $I \subseteq \mathbb{R}$.

- a is a local maximum of f if there exists $\delta > 0$ such that

$$f(a) \geq f(x) \quad \forall x \in I \cap (a - \delta, a + \delta)$$

- a is a local minimum of f if there exists $\delta > 0$ such that

$$f(a) \leq f(x) \quad \forall x \in I \cap (a - \delta, a + \delta)$$

- a is a stationary point of f if f is differentiable at a and

$$f'(a) = 0.$$

Theorem 5.6

Stationary point theorem.

If f has a local maximum or minimum at a and f is differentiable at a, then a is a stationary point of f, i.e.,

$$f'(a) = 0.$$

Proof. Suppose f has a local maximum at a. Then there exists $\delta > 0$ such that

$$f(a) \geq f(x) \quad \forall x \in I \cap (a - \delta, a + \delta).$$

Let h be such that $a + h \in I \cap (a - \delta, a + \delta)$. Then

$$f(a) \geq f(a + h).$$

Dividing both sides by h (for $h \neq 0$) gives

$$\frac{f(a+h) - f(a)}{h} \leq 0 \quad \text{if } h > 0, \qquad \frac{f(a+h) - f(a)}{h} \geq 0 \quad \text{if } h < 0.$$

Taking the limit as $h \to 0$, we get

$$f'(a) \leq 0 \quad \text{and} \quad f'(a) \geq 0,$$

which implies $f'(a) = 0$. The case of a local minimum is similar. □

5.2.2 Rolle's Theorem

Theorem 5.7

Let $a < b$ be real numbers and $f\colon [a, b] \to \mathbb{R}$ be

- continuous on $[a, b]$,
- differentiable on (a, b),
- $f(a) = f(b)$.

Then there exists $c \in (a, b)$ such that

$$f'(c) = 0.$$

Remark

Is quite obvious because the continuity of f ensures that the function must have a maximum or minimum in the interval $[a, b]$, to come back to the same value at b, there must be a point where the derivative is zero.

Proof. By EVT,

$$\exists\, c, d \in (a, b) \text{ such that } f(c) \geq f(x) \geq f(d) \quad \forall x \in [a, b].$$

- if $f(c) = f(d)$, constant function, then

$$f'(x) = 0 \quad \text{for all } x \in (a, b).$$

- if $f(c) > f(a) = f(b)$, $c \in (a, b)$, then

$$f'(c) = 0$$

by the stationary point theorem.

- if $f(d) < f(a) = f(b)$, $d \in (a, b)$, then

$$f'(d) = 0$$

by the stationary point theorem.

□

5.2.3 The mean value theorem

Theorem 5.8

Let $a < b$ be real numbers and $f\colon [a, b] \to \mathbb{R}$ be

- continuous on $[a, b]$,
- differentiable on (a, b).

Then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Remark

The mean value theorem generalizes Rolle's theorem. If $f(a) = f(b)$, then the mean value theorem guarantees the existence of a point $c \in (a, b)$ where $f'(c) = 0$, which is exactly Rolle's theorem.

The idea is we have a tangent parallel to the secant line connecting the endpoints of the function on the interval $[a, b]$. If you move the secant line connecting the endpoints, the tangent line at some point will be parallel to it, which corresponds to the point guaranteed by the mean value theorem.

Proof. Consider the function

$$g(x) = f(x) - f(a) - m(x - a),$$

We have

$$g(a) = f(a) - f(a) - m(a - a) = 0,$$

$$g(b) = f(b) - f(a) - m(b - a) = f(b) - f(a) - \frac{f(b) - f(a)}{b - a}(b - a) = 0.$$

By Rolle's theorem, there exists $c \in (a, b)$ such that $g'(c) = 0$. But

$$g'(x) = f'(x) - m = f'(x) - \frac{f(b) - f(a)}{b - a}$$

So

$$g'(c) = f'(c) - \frac{f(b) - f(a)}{b - a} = 0$$
$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

□

Theorem 5.9

Let $f\colon [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b).

- if $f'(x) > 0$ throughout $x \in (a, b)$, then f is strictly increasing on $[a, b]$,
- if $f'(x) < 0$ throughout $x \in (a, b)$, then f is strictly decreasing.
- if $f'(x) = 0$ throughout $x \in (a, b)$, then f is constant.
- if $f'(x) \geq 0$ throughout $x \in (a, b)$, then f is non-decreasing,
- if $f'(x) \leq 0$ throughout $x \in (a, b)$, then f is non-increasing.

Proof. Only proof $f'(x) > 0$ case, the others are similar.
Let $x_1, x_2 \in [a, b]$ with $x_1 < x_2$. By the mean value theorem, there exists $c \in (x_1, x_2)$ such that

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1).$$

Since $f'(c) > 0$ and $x_2 - x_1 > 0$, we have

$$f(x_2) - f(x_1) > 0 \implies f(x_2) > f(x_1).$$

This shows that f is strictly increasing on $[a, b]$. □

Theorem 5.10

The mean value inequality

Let $f\colon [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). If $|f'(x)| \leq M$ throughout $x \in (a, b)$, then

$$|f(b) - f(a)| \leq M|b - a|.$$

Proof. By the mean value theorem, there exists $c \in (a, b)$ such that

$$f(b) - f(a) = f'(c)(b - a).$$

Taking absolute values, we get

$$|f(b) - f(a)| = |f'(c)||b - a| \leq M|b - a|.$$

□

5.2.4 The second derivative test

Definition 5.4

The second derivative of a function f is the derivative of its first derivative f', denoted by f''. It provides information about the concavity of the function.

Theorem 5.11

Let I be an open interval and $f\colon I \to \mathbb{R}$ be twice differentiable.

- If $f''(a) > 0$, then a is a local minimum on I.
- If $f''(a) < 0$, then a is a local maximum on I.

If $f''(a) = 0$, then the test is inconclusive.

Remark

The concavity of a function can be determined by the sign of its second derivative.

- If $f''(x) > 0$ for all $x \in I$, then f is concave up on I.
- If $f''(x) < 0$ for all $x \in I$, then f is concave down on I.

5.3 Taylor Theorem

5.3.1 Taylor Polynomial

Notation

Write $f^{(n)}$ for the n-th derivative of f.

Definition 5.5

Let $f\colon I \to \mathbb{R}$ be n-times differentiable at $a \in I$. The n-th order Taylor polynomial of f at a is

$$P_n^{f,a}(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k$$

we use $0! = 1$ by convention.

Remark

The thing we cirous about is how accurate the Taylor polynomial $P_n^{f,a}(x)$ approximates the function $f(x)$ near a.

$$\left|f(x) - P_n^{f,a}(x)\right| \leq ?$$

5.3.2 Taylor's Theorem

Theorem 5.12

Let $f\colon I \to \mathbb{R}$ be $(n+1)$-times differentiable at $a \in I$. Then for each $x \in I$, there exists ξ between a and x such that

$$f(x) = P_n^{f,a}(x) + R_n^{f,a}(x),$$

where

$$R_n^{f,a}(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-a)^{n+1}.$$

Example

Let $f(x) = \sin x$, find

$$\left|\sin x - P_3^{f,0}(x)\right|$$

By Taylor's theorem, there exists ξ between 0 and x such that

$$\begin{aligned}
\left|\sin x - P_3^{f,0}(x)\right| &= \left|\sin x - P_4^{f,0}(x)\right| \\
&= \left|R_4^{f,0}(x)\right| \\
&= \left|\frac{\sin^{(5)}(\xi)}{5!}x^5\right| \\
&= \left|\frac{\cos\xi}{120}x^5\right| \leq \frac{|x|^5}{120}.
\end{aligned}$$